Type MS · Explain where bile comes from and how it aids fat digestion

Answer 1 of 9 · Question 2 part a.iii · 3 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

2	(a)	(i)	• A = gall bladder; • B = liver; • C = oesophagus; • D = pancreas;	C- accept pancrease	4
		(ii)	bile;		1
		(iii)	• emulsifies fats/lipids/breaks into small droplets; • increases surface area; • for enzyme/lipase (to act); • neutralises (stomach) acid;		3
	(b)	(i)	any four from • pancreatic juice; • (into th)e small intestine/duodenum; • protease digests/breaks down proteins; • amylase digests/breaks down starch; • lipase digests/breaks down lipids; • (into) maltose/amino acid/glycerol/fatty acid;	If they do not get mp3, mp4, or mp5 They can get a mark for enzyme/named enzyme	4
		(ii)	• insulin; • glucagon;		2
					Total = 14

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 1 of 9 · Question 2 part a.iii · 3 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 2 of 9 · Question 7 part c · 5 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

Number				
7	(a)	$\frac{14}{100}$ x 250 = 35;	Ecf one mark for correct answer from wrong numbers	1
		35 ÷ 10 = 3.5	Full marks for correct final answer of 3.5	1
	(b) (i)	4 x 3.5 = 14 (units);	ecf from 7a	1
		14 – 3 = 11 (hours);	Full marks for correct final answer of 11 hours	1
	(ii)	Any two from: • alcohol is a depressant; • (alcohol) affects the nervous system/brain; • slows reactions/increases reaction time;		Max 2
	(c)	Any five from: • less/no bile produced; • less/no emulsification of fat; • surface area of fat remains low; • less fat digested/broken down (by lipase); • pH of small intestine remains low/acidic; • enzymes denatured/reduced enzyme activity; • reduced/no digestion by protease / carbohydrase/fewer proteins/carbohydrates digested/broken down; • more fat egested;	Allow trypsin for protease and amylase for carbohydrase	Max 5

Total question 7 = 11 marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 2 of 9 · Question 7 part c · 5 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 3 of 9 · Question 2 part a.iii · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

2 (a) (i)	C; (lipase)		1
(ii)	A; (carbon, hydrogen, oxygen)		1
(iii)	C; (liver)		1
(b)	• show effect of enzyme (1) • show that bile salts on their own don't break down fat (1)	Ignore control	2
(c)	• create alkaline/optimum pH (1) • for enzyme to work quickly (1)	Allow increases/adjusts pH	2
(d)	• increase rate of digestion (1) • with no bile salts change occurred after 15 minutes/digested more slowly in tube A (1) • occurred after 5 minutes with bile salts present/digested more quickly in tube B (1)		3

Total 10 marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 3 of 9 · Question 2 part a.iii · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 4 of 9 · Question 2 part d · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

2 (a) (i)	C; (lipase)		1
(ii)	A; (carbon, hydrogen, oxygen)		1
(iii)	C; (liver)		1
(b)	• show effect of enzyme (1) • show that bile salts on their own don't break down fat (1)	Ignore control	2
(c)	• create alkaline/optimum pH (1) • for enzyme to work quickly (1)	Allow increases/adjusts pH	2
(d)	• increase rate of digestion (1) • with no bile salts change occurred after 15 minutes/digested more slowly in tube A (1) • occurred after 5 minutes with bile salts present/digested more quickly in tube B (1)		3

Total 10 marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 4 of 9 · Question 2 part d · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 5 of 9 · Question 3 part a.i · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
3	(a)	(i)	B: A, C and D are not correct as bile is not produced in the gall bladder, small intestine or stomach.		1
		(ii)	Three from: • emulsifies/emulsification fat/lipid; • increases surface area (of fat/lipid); • for lipases/enzymes to digest/work; • neutralises the contents from the stomach;		3
	(b)	(i)	• it contains starch; • it contains protein; • it does not contain reducing sugars/glucose;		3
		(ii)	• contains amylase/carbohydrase; • digested/breaks down the starch; • into glucose/maltose;		3
	(c)		Five from: • peristalsis; • of the food bolus; • (circular) muscles (behind the bolus) contract to push it along; • fibre is indigestible/not broken down; • adds bulk; • stimulates muscles to contract;		5

Total 15 marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 5 of 9 · Question 3 part a.i · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 6 of 9 · Question 3 part a.ii · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
3	(a)	(i)	B: A, C and D are not correct as bile is not produced in the gall bladder, small intestine or stomach.		1
		(ii)	Three from: • emulsifies/emulsification fat/lipid; • increases surface area (of fat/lipid); • for lipases/enzymes to digest/work; • neutralises the contents from the stomach;		3
	(b)	(i)	• it contains starch; • it contains protein; • it does not contain reducing sugars/glucose;		3
		(ii)	• contains amylase/carbohydrase; • digested/breaks down the starch; • into glucose/maltose;		3
	(c)		Five from: • peristalsis; • of the food bolus; • (circular) muscles (behind the bolus) contract to push it along; • fibre is indigestible/not broken down; • adds bulk; • stimulates muscles to contract;		5

Total 15 marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 6 of 9 · Question 3 part a.ii · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 7 of 9 · Question 7 part c · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

7 (a)	(i)	• volume/concentration of enzyme/lipase; • volume/concentration of indicator; • volume/same type of olive oil; • volume/mass of bile saltes	Ignore amount for all mps	3
	(ii)	40°C;		1
	(iii)	no values given between 30°C and 40°C; no values given between 40°C and 50°C; optimum could be between temperature intervals given/below/above 40ºC/50ºC;		3
(b)	(i)	• tube 3 = 11.5 hours and tube 8 = 24 hours; • 24 - 11.5 = 12.5 hours;	Ecf incorrect values in a subtraction calculation with a correct answer 1 mark	2
	(ii)	• lipase digests/breaks down olive oil; • produces fatty acids; • these acids lower the pH;		3
(c)		• no bile salts so no emulsification of oil/oil not broken down to droplets; • less surface area (for lipase to work on); • lipase only working very slowly/not very active/takes longer to digest oil;		3

Total 15 marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 7 of 9 · Question 7 part c · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

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Type MS · Explain where bile comes from and how it aids fat digestion

Answer 8 of 9 · Question 3 part b · 2 marks

2024 JUN · Paper 02 · find part b in this mark scheme extract

Number			
3 (a)	Smoking cigarettes; Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ; OR lack of exercise; obesity/heart disease/high blood pressure/ type ii diabetes/eq;	Allow drugs or named drug and their effect on health for 2 marks.	(2)
(b)	Any two from: • production of bile reduced/affected; • less fat emulsified/fat not emulsified; • lipase less active/lipase unable to digest as much (fat)/slower lipase activity;	Allow bile not produced Allow description of emulsification	(2)
(c)	blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina;		(1)
(d)	cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata);	Allow any part of the brain that is affected by alcohol	(1)
(e)	Any three from: • reduces release of neurotransmitters; • reduced diffusion across synapse; • less neurotransmitters bind to receptors on post-synaptic membrane; • slows/ fewer transmission of impulses; • slower reaction /reflexes;		(3)
(f)	Any two from • increased heart rate; • increased breathing rate; • pupil dilation; • increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose; • increase in blood flow to brain/muscles; • increase in (aerobic) respiration (in muscles);		(2)

Total for Question 3 = 11 marks

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 9 of 9 · Question 2 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

2	(a)	(i)	• A = gall bladder; • B = liver; • C = oesophagus; • D = pancreas;	C- accept pancrease	4
		(ii)	bile;		1
		(iii)	• emulsifies fats/lipids/breaks into small droplets; • increases surface area; • for enzyme/lipase (to act); • neutralises (stomach) acid;		3
	(b)	(i)	any four from • pancreatic juice; • (into th)e small intestine/duodenum; • protease digests/breaks down proteins; • amylase digests/breaks down starch; • lipase digests/breaks down lipids; • (into) maltose/amino acid/glycerol/fatty acid;	If they do not get mp3, mp4, or mp5 They can get a mark for enzyme/named enzyme	4
		(ii)	• insulin; • glucagon;		2
					Total = 14

Type MS · Explain where bile comes from and how it aids fat digestion

Answer 9 of 9 · Question 2 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks